Reproduction in plants

Many plants reproduce sexually. In plants *pollen* is the male sex cell and *ovum* (plural *ova*) is the female sex cell. Flowers are the organs of sexual reproduction in angiosperms (flowering plants). In sexual reproduction male and female sex cells must come together. This process in plants is called *pollination*.



Flowers are highly modified shoots made up of four major parts. *Sepals*, modified leaves, make up the outer series. The next inner series, the *petals*, are also modified leaves which are usually not green but brightly coloured. The third series consists of the *stamens*, the pollen-producing male portion of the flower. A stamen is made up of an *anther* and a *filament*. The innermost series are the *carpels*, female structures that produce the seeds. The *stigma* is the top of the carpel and is sticky to catch pollen. Carpels can join to form a structure called the *pistil*. The seeds of angiosperms develop in an *ovary*, a part of the carpel that surrounds and protects the egg-containing *ovules*. Seeds develop from the ovules after pollination and fertilisation of the eggs.

Complete the missing parts of the table below. The first one has been done for you.

Flower part	What it does
stamen	male part of the plant, produces pollen
anther	**1**
2	male reproductive cells
carpel	**3**
4	inside the carpel, contains ovules
stigma	**5**
6	develop into seeds after fertilisation
petals	**7**
8	outer covering or bud

Pollen from one flower is transferred to the stigma of another flower. The pollen can be carried on the bodies of insects or by the wind. Insect-pollinated plants have brightly coloured and scented petals. Wind-pollinated flowers are small, often green and unscented with the anthers and stigmas dangling out of the flower.

9 Give reasons why insect-pollinated plants have brightly coloured and scented flowers.

The pollen joins with an ovum to produce a seed. Seeds contain all the information to grow into a new plant. Seeds need to be spread far away from the parent plant. They can be scattered by the wind, carried in animal fur, eaten and passed out in droppings, some are even dispersed by the fruit exploding. They can also be planted on purpose by humans.

10 Explain why the anthers and stigmas of wind-pollinated plants dangle out of the flower.

11 Give three means by which seeds are moved from the parent plant.

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Types of pollination

Pollination is the transfer of pollen from a stamen (male structure of a flower) to a stigma (opening) of the pistil (female structure). The transfer can occur within the same flower (self-pollination) or to a different flower (cross-pollination) of the same species. Different flowers have evolved in different ways to ensure pollination takes place.

12 What are *angiosperms*?

Self-pollination is less dependent on outside forces and more reliable than cross-pollination. However, self-pollination does not create variation between offspring. Some species establish themselves in an area by repeating many copies of the parent plant. If all the offspring are identical, then they run the risk of all being destroyed if their environment changes.

13 Define *environment*.

Seeds that have genetic material from two different plants are more likely to produce more varied offspring that are better able to cope with changes. Some plants have developed ways to make self-pollination less likely. Some plants have 'male' and 'female' versions, with only the male part of flowers on some plants and the female part of flowers on others. In some other types of plants the pollen is produced at a time when the pistil is not ready to receive it.

14 Identify the main male structure of a flower.

15 Identify the main female structure of a flower.

16 Which produces the greater variety of offspring, cross-pollination or self-pollination?

17 Describe two ways in which plants try to stop self-pollination.

Vast amounts of pollen are carried through the air from plants using cross-pollination for fertilisation. Insects, birds and bats are also involved in cross-pollination. They tend to go between flowers of the same species of plant in search of nectar. Bees are especially important in pollination. Often beehives are placed in orchards to ensure that pollination takes place. People are also directly responsible for pollination. Some plant producers hand-pollinate specific plants to eliminate the randomness of wind or insect pollination.

Some people are allergic to air-borne pollen. They sneeze, have inflamed nose and eye membranes and suffer difficulties in breathing.

18 Give a reason why so much pollen is produced by some plants.

19 Name two ways used by plants to attract pollinating creatures.

20 Describe some features of bees that might make them good pollinators.

21 Moths fly at night and find their mates by scent. Give two properties of a plant that will tend to be pollinated by moths.

22 Give a reason why breeders growing plants in glasshouses will often release a bee into the enclosed structure.

23 Which disease is described by the allergic reaction mentioned above?

24 In Mexico and Central America bees and hummingbirds pollinate vanilla flowers, but in Australia, pollination is done by hand. Give two possible reasons for this.

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